From Connection Loss to Collision: Closed-Loop Fault Assessment of a Fly-Connectome SNN

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***Abstract*—We assess how connection loss affects action timing and first contact in a fly-connectome spiking network, using 576 test episodes organized into paired campaigns under two frozen feedback laws. Both laws complete 48/48 healthy RF-0 tasks. At 50% connection-row deletion, 1-s proportional/relay success is 37/48 versus 48/48; the prespecified secondary net-loss contrast is -22.9 pp (95% whole-seed CI $[-33.3, -12.5]$). Separately recalibrated, same-prior RF-A/B each yield an uncertain -4.2 pp $[-12.5, 0]$. Unequal healthy reserve, recalibration and test cohorts limit attribution. Post-hoc analysis finds later mean triggers at 50% deletion, yet identical schedules yield law-discordant contacts. Among 543 event-free burst completions with follow-up, subsequent command features are zero: late outcomes occur during an observed walking regime. A frozen command warning detects only 2/13 RF-0 contacts; an exact-count definition stabilizes its boundary semantics. An executable evidence interface checks pairs, clocks and labels, with independent replay and reuse on a delayed tank. Together these measurements resolve stages of finite-horizon preservation.**

***Index Terms*—spiking neural networks, fault assessment, closed-loop control, reproducibility**

I. Introduction and Positioning

How does matched connection loss in a connectome-derived spiking neural network (SNN) affect first contact as feedback, observation horizon and calibrated sensing change? We pair healthy and damaged runs, resolving neural commands, action timing and contact separately.

Shiu et al. [1] establish connectome-derived LIF sensorimotor predictions; here outputs drive a synthetic pursuit–escape plant. An [2] studies task/state-dependent preservation under sparsification, and Jo [3] couples damaged circuits to virtual-arena behavior. We measure paired geometric first contact under two frozen laws and a declared horizon, and predict subsequent contact from a 400-ms prefix. Task dependence and neural/behavioral dissociation themselves are not new claims.

DriveFI searches closed-loop safety violations [4]; SpikeFI, SpikingJET and enpheeph provide SNN fault injection [5]–[7]. Common random numbers [8], trajectory monitoring [9], benchmarking [10], [11] and physical fault propagation [12] are established. Our interface complements these methods with executable pairing, timing and native-replay obligations.

We contribute: (i) stage-resolved paired timing/contact measurements, including law-discordant outcomes under shared schedules and horizon-sensitive, uncertain RF transfer; (ii) frozen-prefix warnings with 11/13 missed RF-0 contacts and cross-configuration false alarms; and (iii) executable pairing, timing and exposure contracts, tested on archived actuator/endpoint errors and a delayed tank. Timing remains descriptive; contracts test realized against declared actuation.

II. Neural Experiment and Statistical Design

A. Network, Feedback and Calibration

The FlyWire v783-derived model [13] has 138,639 LIF neurons and 15,091,983 signed, contact-weighted connection rows; cell labels use FlyWire annotations [14]. Shiu parameters [1] run in Brian 2 [15], with 0.1-ms steps, 2.2-ms sensory refractoriness and state/weight restoration per episode. Assumed RFs stimulate 104 LC4 and 210 LPLC2 cells; eight descending cells provide command features (Fig. 1(a–c)).

Between events, $\dot{v} = (v_0 - v + g)/(20\,\mathrm{ms})$ and $\dot{g} = -g/(5\,\mathrm{ms})$. At -45 mV, reset $(v, g) = (v_0, 0)$, $v_0 = -52$ mV; both derivatives pause during refractoriness. Recurrent events add 0.275 mV per signed contact to g after 1.8 ms; external events add 68.75 mV to v.

Input starts at 100 ms with visual period $\delta_\mathrm{v} = 0.020$ s. In degrees, β is world bearing minus heading, $a = (180/\pi)\arctan(0.05/d)$ is half-angle at separation d, $\sigma = \max(30^\circ, a)$ and $\ell = (a - a_\mathrm{prev})/\delta_\mathrm{v}$:

$$\begin{aligned} u &= \mathrm{clip}_{[0,1]}[\ell_+ e^{-a/60}/150], \\ \lambda_i &= \mathrm{clip}_{[0,150]}[150 w_i u e^{-[\mathrm{wrap}(\phi_i-\beta)^2+\psi_i^2]/(2\sigma^2)}]\ \mathrm{Hz}. \end{aligned} \tag{1}$$

Pre-cue $u = 0$ masks the initial analytic expansion; the 100-ms input uses 80–100-ms angle history. Wrap maps to $[-180^\circ, 180^\circ)$. RF azimuth/elevation ϕ_i, ψ_i set $w_i = 1$ for LC4 and $1 + 0.8(\psi_i + 30)/90$ for LPLC2; other external rates are zero. A 0.3-m/s pursuer starts 0.25 m away at $\pm 30^\circ$ and targets the agent's segment-start position. Linear segments use starting headings; agent heading updates at segment end, capped at $\pm 100^\circ$. Decisions apply next frame. First trigger starts a heading-fixed 0.35-m/s burst 250 ms later for 450 ms; otherwise speed is 0.05 m/s. Actuator events split segments; first contact is solved within them at 0.05-m center separation. Failure latches; resources continue to $H = 1$ s; warnings use pre-contact data.

Per map, 20 passive development episodes (four seeds, five bearings in $[-60^\circ, 60^\circ]$) fit affine score z. Standardized eight-cell five-frame means over 200–400 ms predict bearing/60° with ridge penalty 10 and an unpenalized intercept. Trigger threshold is half the development tenth percentile of 200–300-ms episode command means, floored at 2.5 Hz.

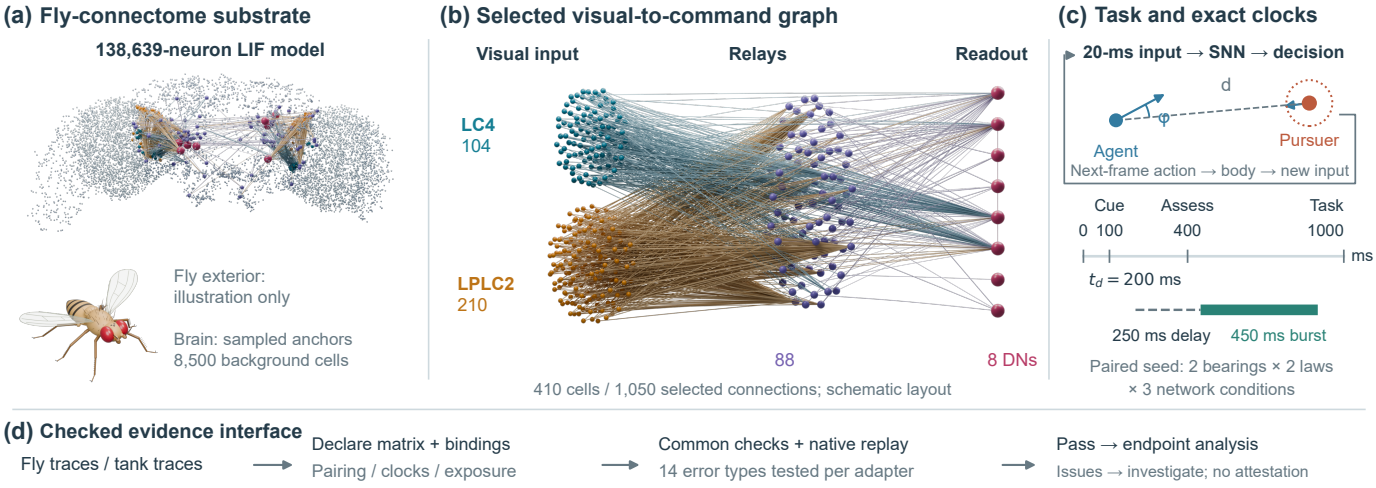


Fig. 1. System and checked assessment interface. (a) Representative model anchors and an illustrative fly; (b) a selected 410-cell/1,050-connection graph. The layout is schematic, not measured neurites or the complete recurrent graph. (c) Sampled feedback and exact burst events; $t_d = 200$ ms is illustrative. (d) Native traces enter common checks and independent replay before endpoint analysis.

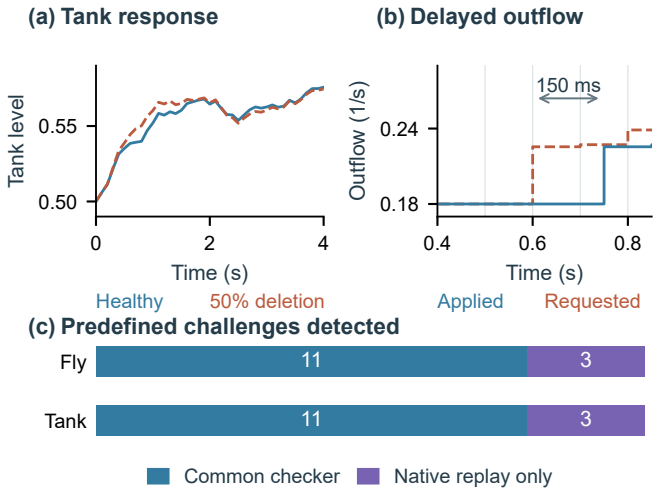


Fig. 2. Cross-plant reuse and predefined error challenges. (a) First registered tank seed, high inflow, proportional law; level axis is zoomed, safe interval $[0.2, 0.8]$. All 96 example tasks remain safe. (b) A requested outflow first changes at 0.60 s and applies at 0.75 s, between 100-ms frame boundaries. (c) Eleven common-checker and three native-only cases are detected per adapter, not a general coverage estimate.

Before cue or below threshold, steering is zero; otherwise P/R apply $-600\,\mathrm{clip}(z,-1,1)/-600\,\mathrm{sgn}(z)$ deg/s outside the burst. Eight held-out validation clusters per RF give RMSE $65.9°/67.8°/67.2°$: z poorly reconstructs closed-loop bearing.

B. Pairing, Cohorts and Endpoints

Shared uniforms give nested deletions $w_j^{(f)} = w_j \mathbf{1}[V_{sj} \geq f]$ and input events $\mathbf{1}[U_{sk\tau i} < \lambda_{ki}\Delta t]$. Indices denote seed s, frame k, neural step τ, cell i and connection row j; $\Delta t = 10^{-4}$ s. Feedback changes λ, while input variates and row masks remain paired. Seeds and hashes are traceable, without cross-platform bit-identity guarantees.

RF-0/A/B use 24/12/12 independent test clusters, each containing both bearings, two laws and healthy/25%/50% deletion: 288/144/144 episodes. New maps independently redraw 314 azimuths from the same hemisphere prior (left 10–90°, right -90–$-10°$), preserving identities, elevation and gradients. Each repeats the frozen development recipe, validates on 32 healthy episodes in eight separate clusters, then freezes thresholds. All maps, seeds and failures remain; no test tuning or efficacy stopping occurs. Transfer includes recalibration, not zero-shot or RF-isolated causality.

With $Y = 1$ meaning no contact through H, registered primary endpoints are paired loss $L = Y_H(1 - Y_F)$, gain $G = (1 - Y_H)Y_F$ and activity exposure. Net loss $D = L - G$ is derived. The relay-minus-proportional contrast and warnings are prespecified secondary; trigger-timing and clearance/heading diagnostics are post-hoc. For cumulative spikes S and $T_* = \min(T_F, T_H)$, own-stop and common-duration ratios are

$$R_{\mathrm{stop}} = \frac{S_F(T_F)}{S_H(T_H)}, \qquad R_{\mathrm{common}} = \frac{S_F(T_*)}{S_H(T_*)}. \tag{2}$$

Here T is contact-frame end or H, at 20-ms resolution. Common duration remains outcome-dependent and does not equate sensory histories; fixed 400-ms/1-s windows are separate endpoints.

Binary loss/gain average both bearings within seed, then seeds. Exposure uses the median of paired ratios, retaining both bearings; warnings pool trial-level confusion counts. Pointwise 95% CIs resample whole seeds 20,000 times (RNG 20261022 for RF-0; 20261023 for transfer), retaining their complete condition matrix and recomputing the corresponding statistic. CIs condition on stored maps/calibrations. Symmetric-null sign-flip tests use separate Holm families for two baseline and four transfer contrasts. Maps and separate angle/action cohorts are not pooled.

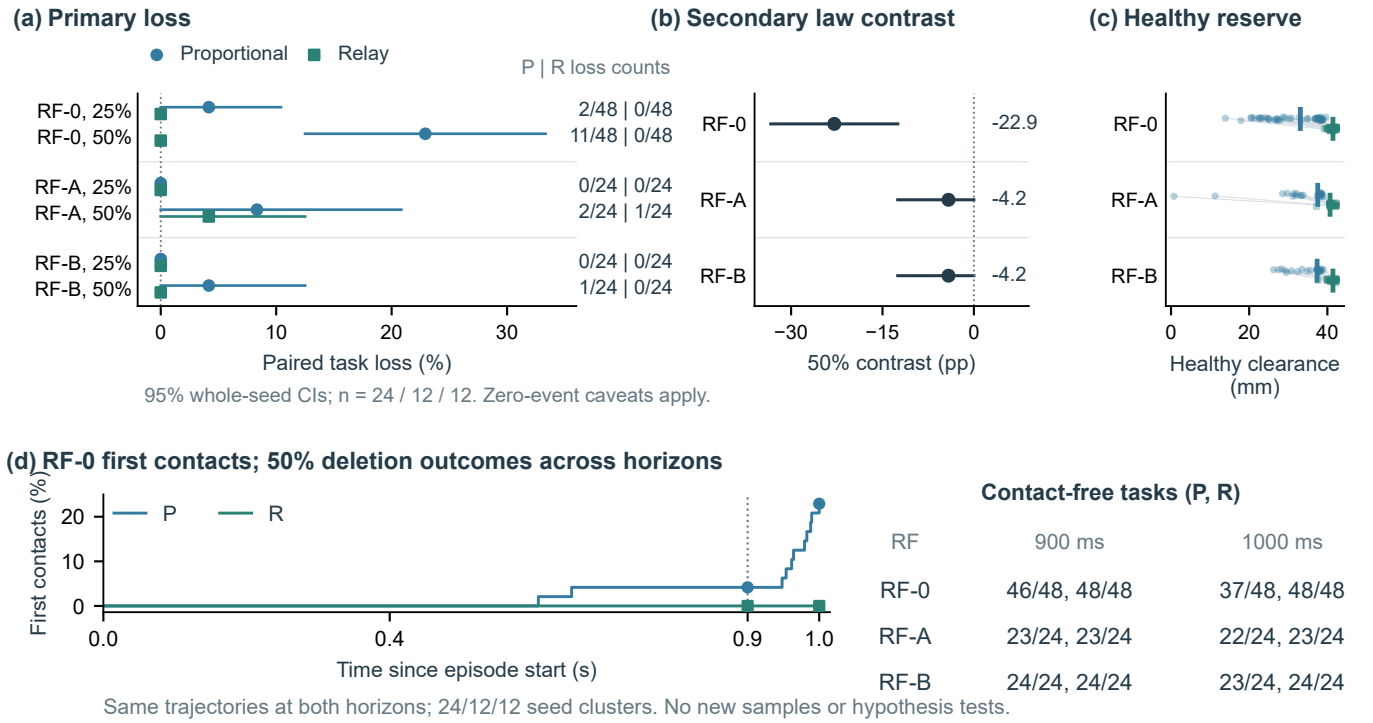


Fig. 3. Paired loss across sensory realizations and horizons. (a) Primary losses and (b) prespecified secondary R-minus-P contrasts (percentage points), with pointwise 95% whole-seed CIs, $n = 24/12/12$, conditional on saved calibrations. All healthy tasks succeed; no gains occur. (c) Paired healthy minimum-clearance points and median marks show unequal reserve. (d) RF-0/50% first contacts and all-map 900/1000-ms success counts (P, R); both endpoints reuse the same 24/12/12 seed clusters, not new replicates or tests. Maps differ in draws, recalibration and cohorts, not an isolated RF effect.

Warnings use only the 400-ms prefix. Command count C sums eight cells across fifteen 20-ms bins in 100–400 ms; saved rates are exact 50-Hz multiples, allowing lossless integer recovery. Each 32-record healthy validation set gives a linear/type-7 fifth percentile: 14/14.55/13.55 counts in RF-0/A/B. The frozen strict rule $C < Q_{.05}$ does not flag ties; displayed $C/2.4$ Hz never determines the label. Activity uses a law/bearing-specific healthy reference and a frozen 95th-percentile absolute-log-ratio cutoff; an upper-only comparator flags ratios above two. Forecast uses exact simulated state and pending actuation, holding the current turn; if untriggered at 400 ms it assumes a 420-ms trigger. Its schedule-only baseline sets future turn to zero. Each RF freezes its fifth-percentile forecast cutoff on healthy validation; these forecasts are not neural-only detectors.

III. Executable Evidence Contract

A campaign declares seed, condition, law, fault, horizon, frame duration and resource unit. Adapters bind native traces to these identities, sources, maps, calibration, input/mask digests, exposure, resources and first failure (Fig. 1(d)). Common checks test these fields and pairing; native replay independently regenerates inputs and plant transitions. Hashes bind bytes, not execution. Shared variates need not yield identical feedback-dependent events; warning prefixes exclude future labels.

Native replay checks 250-ms trigger-to-onset delay, 450-ms burst duration and interval-consistent horizon flags. In 72 separate historical records it finds 17 delays of 260 ms, four completed bursts of 440 ms and six incorrect horizon flags. These overlapping cases are not pooled with current cohorts: nominal parameters would misstate fast-motion duration or endpoint actuation. Common-schema findings retain row indices; native semantics remain adapter obligations.

The tank obeys $\dot{h} = I - u - 0.04h$, $h(0) = 0.5$, safe interval $[0.2, 0.8]$ (Fig. 2(a)). Eight seeds $\times$ two inflows $\times$ two laws $\times$ three sensor-deletion levels give 96 examples; deleted samples hold their last value. Analytic replay checks accounting and 150-ms actuation on a 100-ms grid (Fig. 2(b)). All remain safe, demonstrating reuse rather than neural tolerance.

Fig. 2(c) reports eleven common-checker and three native-only cases per adapter, not general coverage. A synthetic zero-neural-readout logger passes input/plant replay without neural dynamics: trace consistency cannot authenticate execution, unlogged state or safety.

IV. Results and Conditional Interpretation

A. First-Contact Loss Depends on the Declared Horizon

Both laws completed 48/48 healthy RF-0 tasks; Fig. 3(a) shows paired losses. Fig. 3(d) rescores the same 50%-deletion trajectories at 900/1000 ms across all three RFs. RF-0 has nine additional P contacts in the final 100 ms (latest 999.962 ms); law-discordant RF-A/B contacts also occur after 900 ms. These

descriptive rescorings add neither samples nor confirmatory tests; RF-0's repeated endpoints were registered.

The prespecified secondary 1-s R-minus-P net-loss contrast (Fig. 3(b)) was -22.9 pp (95% CI $[-33.3, -12.5]$; Holm $p = 0.00195$). RF-A/B each yielded -4.2 pp, CI $[-12.5, 0]$; all four new map/dose Holm p values are 1. Each new 50% contrast has only one nonzero cluster. These estimates establish neither equivalence nor an isolated RF effect; no between-map test is claimed. The baseline 25% contrast remains uncertain ($p = 0.50$). No 25% loss occurs in A/B; twelve event-free clusters still allow a one-sided 95% upper bound of 22.1% for any loss in a new identically defined two-bearing cluster.

All 576 records triggered. Post-hoc healthy–50%-deletion delays averaged 80.0/57.5/57.5 ms in RF-0/A/B (SD 42.9/46.6/29.7 ms across 48/24/24 bearing pairs; 24/12/12 independent seeds, P/R duplicates counted once). Trigger means first post-cue crossing by the eight-cell five-frame mean at 20-ms resolution. All 288 P/R pairs shared trigger/burst schedules under pre-trigger gating, yet 15 had different contact outcomes. Both RF-0 25% P contacts occurred without delay: timing alone neither determines outcome across laws nor accounts for every contact.

Among 543 contact-free burst completions with follow-up, all 3,114 decisions at/after burst end and before contact/H had zero eight-cell features and gated turn; 474 records retained whole-network spikes. Their 13 later contacts occurred during this observed walking regime. This post-hoc stage result contextualizes burst-end geometry and finite follow-up; it is not a universal silence rule or an isolated delay mechanism. Four prior contacts, 17 ongoing bursts and 12 zero-follow-up records remain outside this risk set.

In Fig. 3(c), paired healthy minimum-clearance points and vertical median marks show unequal reserve: RF-0 P/R medians were 33.1/41.4 mm (400-ms absolute heading $88.3°/100°$). Reserve, burst-end state and late follow-up are possible contributors, not isolated mechanisms; RF calibration and cohorts also vary together.

B. Frozen Scalar Warnings Miss Subsequent Contacts

Across healthy/25%/50% conditions, the command warning detected 2/13 RF-0 contacts, 2/3 in A and 0/1 in B (Fig. 4(a)); successful-task false alarms were 51/275, 42/141 and 24/143 (Fig. 4(b)). Exact-count re-scoring preserves all historical labels and CIs; rate-first rounding would falsely flag 19 RF-0 successes tied at 14 counts. The upper-only activity rule detected none. Exact-state forecasts detected all observed contacts but falsely flagged 57/275, 25/141 and 33/143 successes; baseline schedule-only false alarms were 84/275. Their geometry/actuation information precludes a neural-only comparison.

RF-0/A/B sensitivities have only 12/2/1 failure clusters. Of 20,000 resamples, 2,217/7,099 lack a positive class in A/B; those draws are omitted only for the corresponding conditional rate. Zero-width empirical intervals do not establish zero population risk. Own-stop/common-duration medians coincide in these campaigns (maximum paired difference below 0.006), so the historical large exposure artifact does not recur. P/50% median 400-ms/1-s activity ratios remain 0.981/1.505, 0.961/1.212 and 0.956/1.139. A separate eight-cluster $\pm 45°$ test flags 22/32 healthy activity records and retains an uncertain law contrast. These results bound the tested frozen criteria, not all neural monitoring.

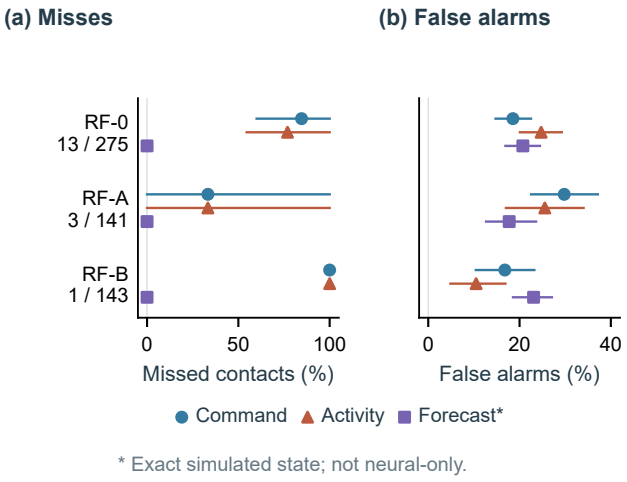


Fig. 4. Frozen secondary warning operating points. Rows state contact/success denominators; RF-0/A/B contain 24/12/12 independent seed clusters. Markers show (a) missed-contact and (b) false-alarm rates with pointwise 95% whole-seed bootstrap CIs. Activity is two-sided; the forecast holds the current command using exact simulated state. Positive-class support is 12/2/1 seed clusters. Zero-width and class-conditional empirical intervals do not establish zero population risk.

V. Scope, Artifacts and Conclusion

Evidence concerns one graph, an assumed RF family, a synthetic plant, unequal healthy reserve and sparse failures. Roughly $66°$ held-out RMSE limits bearing reconstruction. Walk-only contact by 0.8 s reflects plant speeds; one burst followed by slower walking makes 1-s success finite-horizon preservation, not sustained avoidance. Functional row loss establishes no biological tolerance, device upset rate, hardware safety or energy/area benefit; trace consistency cannot authenticate neural execution.

Generating data [16], schema/horizon analysis [17] and V10.1 timing [18] retain distinct provenance. The fixed implementation [19] supplies `verify_v10_1.py`, timing/stage entries, count re-scores, tests and Fig. 4 data; `PUBLICATION.md` binds this PDF. Generation metadata remain unchanged. Only fresh neural runs need external hash-matched matrices; unresolved redistribution terms are respected.

Connection loss shifted mean triggers later, while shared schedules yielded law-discordant contacts. Stage analysis locates late outcomes in an observed zero-command walking regime; healthy reserve and delay contributions remain unisolated. Exact-count warnings retain task-label errors. The checked interface makes pairs and clocks testable across two plant schemas. This task-level assessment provides a basis for evaluating how future device-level nonidealities propagate through neuromorphic controllers to closed-loop behavior.

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